

Qatar-1b

R-0229

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_181015 · created 2026-07-17T03:25:13+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458406.7383 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.055
Transit depth (Rp/Rs)^2	3.07 +/- 1.92 [%]
Orbital Inclination (inc)	88.91 +/- 2.87
Airmass coefficient 1 (a1)	0.2 +/- 0.13
Airmass coefficient 2 (a2)	0.84 +/- 0.4
Scatter in the residuals of the lightcurve fit is	64.47 %
Best Comparison Star	#2 - [590, 161]
Optimal Aperture	0.98
Optimal Annulus	7.59
Transit Duration (day)	0.0819 +/- 0.0071

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.055 vs 0.1463 (deviation 0.38σ)
Duration fit vs published	0.0819 d vs 0.0692 d (deviation 1.29σ)
Mid-time O–C	-3.2 min
Depth SNR	2.3
Residual scatter	64.47 %

Light curve

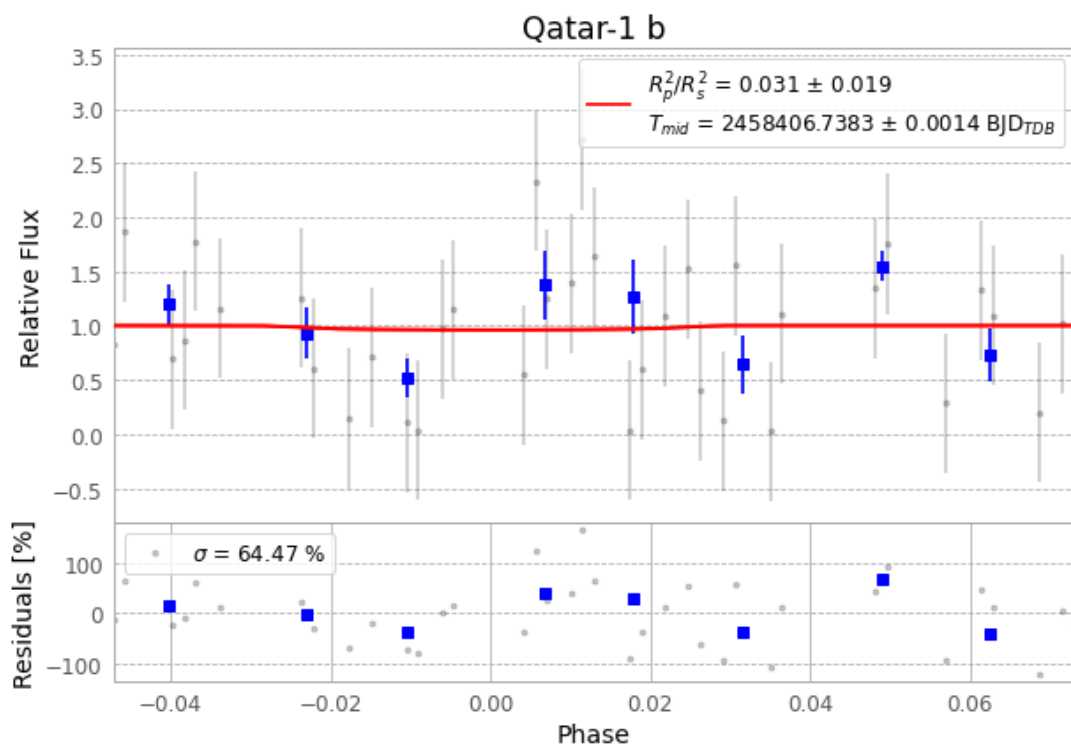


Figure 1 — FinalLightCurve_Qatar-1 b_2018-10-14.png (SHA-256 14cf54bf54aa9c29...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [537, 109], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 76b27d91a78b7e62...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

84 FITS frames (55 MB), Qatar-1181015040012.FITS → Qatar-1181015080912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-181015.log (1630b9953bca...), FinalLightCurve_Qatar-1 b_2018-10-14.png (14cf54bf54aa...), FinalLightCurve_Qatar-1 b_2018-10-14.csv (c94cea112aff...)

Compute resources

Wall time 245.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 03:25Z.